Issue 4

RESCUER Newsletter

March 2026

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The Doctoral Network **RESCUER**

(Resilient Solutions for Coastal, Urban, Estuarine and Riverine Environments)

aims to train 10 young researchers to tackle medium and long term water challenges that face coastal communities, focusing on forecasting and modeling of coastal, riverine, and urban flooding and associated water quality issues.

EDITORIAL

Welcome to the fourth issue of the RESCUER project newsletter!

The past months have once again been marked by important achievements, dynamic collaboration, and continued progress across the consortium. During this period, the network successfully delivered **Workshop 4: Code Development Tools & High-Performance Computing**, held online in November 2025, followed by **Workshop 5**, hosted in Coimbra, Portugal, in March 2026. These training events provided fellows with valuable expertise in scientific programming, experimental methods, hydraulic modelling, and advanced computational tools.

This edition highlights our recent milestones and preparations for upcoming activities You will also discover research updates from our fellows and key moments from recent events across the network.

https://www.rescuer-msca.net/.

Stay tuned and happy reading!

Funded by the European Union

The project has received funding from the European Union's Horizon Europe research and innovation program under the grant agreement No. 101119437

RESCUER's 4th Workshop

The **RESCUER Doctoral Network** successfully organized **Workshop 4: Code Development Tools & High-Performance Computing**, held online on **19–20 November 2025**. This two-day training event brought together RESCUER fellows and participants for an intensive program dedicated to modern programming practices and advanced computational tools for environmental modeling.

This virtual training event offered 12 hours of instruction covering collaborative software development on Github and GPU Programming.The workshop was organized by SUEZ-RPT via Microsoft Teams.

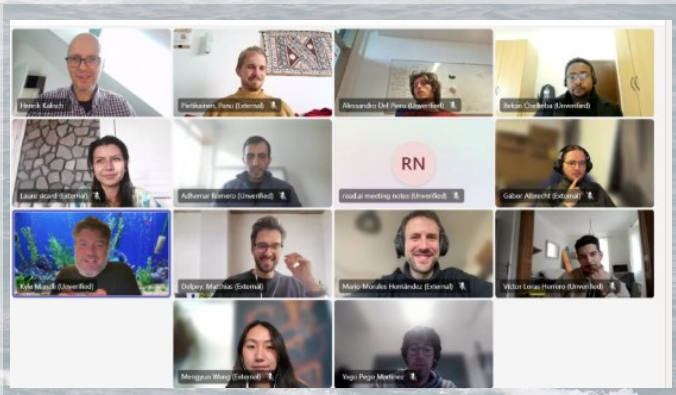
As numerical simulation becomes an increasingly important tool for understanding and predicting coastal, urban, estuarine, and riverine flooding processes, strong expertise in scientific programming and high-performance computing is essential for early-stage researchers. In this context, Workshop 4 was designed to strengthen participants' technical skills in modern code development practices and advanced computational techniques used in large-scale environmental modeling.

Throughout the workshop, participants explored topics such as structured scientific programming, collaborative software development, reproducibility, efficient numerical implementation, parallel computing, and the use of HPC infrastructures to accelerate complex simulations.
Participants benefited from expert lectures delivered by two internationally recognized speakers:

- Kyle Mandli (NSF-NCAR, USA) who shared valuable experience on scientific software development and computational modeling practices.
- Mario Morales Hernández (University of Zaragoza), who presented advanced perspectives on numerical simulation tools and high-performance computing for hydraulic and environmental applications.

Their expertise offered participants both theoretical foundations and practical guidance directly relevant to current research challenges.

Beyond the technical content, the workshop also provided an important opportunity for interaction and exchange among fellows across the RESCUER network, reinforcing collaboration within the consortium and fostering a shared culture of open, robust, and efficient scientific computing.



Online meeting during Workshop 4

Workshop 4 is part of the broader RESCUER training programme, which combines scientific excellence, interdisciplinary collaboration, and transferable skills development to prepare a new generation of researchers capable of addressing future water-related risks and resilience challenges.

We warmly thank the speakers, organizers, and all participants for making this workshop a success.

RESCUER's 5th Workshop

RESCUER Workshop 5: Hands-On Training in Experimental Methods and Multi-Physics Simulation

From **23 to 27 March 2026**, the **RESCUER Doctoral Network** successfully held its fifth training workshop in **Coimbra, Portugal**, bringing together fellows, supervisors, and researchers for an intensive week of scientific training, practical activities, and networking.

Hosted by the University of Coimbra, the workshop offered a rich interdisciplinary program combining **experimental hydraulics, environmental monitoring, field observations, and advanced computational coupling techniques**. The event reflected RESCUER's commitment to providing doctoral candidates with broad and complementary skills to address future water-related challenges.

The first part of the workshop focused on **chemical and physical experimental methods**, including lectures and laboratory sessions delivered by Rui Martins and Rita Carvalho. Participants explored water chemical characterization, hydraulic physical modelling, and **Particle Image Velocimetry (PIV)** techniques, combining theoretical foundations with hands-on laboratory testing and student presentations.

A fieldwork day in Figueira da Foz allowed fellows to connect theory with real environmental systems through estuary observations and coastal site visits. These activities provided valuable insight into natural hydrodynamic processes, monitoring strategies, and the practical challenges of environmental engineering in real-world settings.

The second part of the workshop was dedicated to **co-simulation and coupled numerical modelling**, led by Claes Eskilsson. Sessions covered MPI fundamentals, scientific libraries, coupling strategies, data mapping, and the use of tools such as **preCICE** and **MUI** for multi-physics simulations. Through lectures and exercises, participants gained practical experience in linking models and improving interoperability between numerical codes.



Hands-on session

Beyond the scientific program, the workshop also offered many opportunities for exchange and collaboration through shared dinners, site visits, and networking moments along the Mondego River and the Portuguese coast. These interactions further strengthened the collaborative spirit of the RESCUER consortium.



The fellows during a laboratory session



Lecture on code coupling

Workshop 5 once again demonstrated the value of RESCUER's training strategy: combining theory, experimentation, computation, and international collaboration to prepare a new generation of researchers capable of tackling complex flooding and environmental resilience challenges.



Social dinner



Visiting Figueira da Foz

We warmly thank the hosts at the University of Coimbra, all lecturers, organizers, and participants for making this workshop an outstanding success.

Focus on University of Zaragoza

by prof. Pilar Garica Navarro and Laure Sicard, PhD Fellow at UNIZAR,



PhD Fellow Laure Sicard

The consortium member University of Zaragoza is a public institution of higher education and research serving society. It is the largest center of higher education in the Ebro Valley, combining a tradition of almost five centuries (it was founded in 1542) with the continuous updating of its academic offerings. Its primary mission is to generate and transmit knowledge with the aim of the holistic development of individuals. It is a high-quality, socially responsible, and open university that aspires to be an instrument of social transformation for economic and cultural development. Research activity is channeled through research groups. At the University of Zaragoza, 2,884 researchers work in 170 research groups. The University of Zaragoza has five of its own university institutes, four joint institutes, and one affiliated institute. The I3A is one of the university institutes where the participating group is located.

The team GHC brings extensive expertise in the mathematical and numerical modeling of hydro-morphological processes at the Earth's surface, such as rainfall-runoff transformation, erosion and sediment transport, fluvial flows and flash floods, flooding, mud and debris flows, etc. This team of researchers and developers is highly qualified to undertake the tasks proposed in the STREAM project, an active and leading group in the formulation, implementation, and testing of computational models. The subproject1.GHC team has extensive experience programming computational codes for these models, using high-performance computing (HPC) techniques that allow numerical solution of physical equations using Graphics Processing Units (GPUs) or split the computation between multiple GPUs. These HPC programming techniques result in a significant reduction in the computational time required for model calculations. This allows for highly efficient 2D computational simulation tools for hydro-morphological and environmental processes at the Earth's surface scale. They also have extensive experience in technology transfer to national and international companies. The models developed by the subproject 1.GHC team are at the forefront of computational approaches for simulating hydrological and morpho-dynamic flows, achieving high quality standards at reduced computational costs.

Research work conducted as part of RESCUER

As part of RESCUER, Universidad de Zaragoza is in charge of the PhD project, entitled *Multiphysics shallow-water models for river and estuary dynamics*. (WP 2)

Rivers and estuaries are complex, interconnected systems which are subject to a multitude of physical, chemical and biological

effects. Proper maintenance includes monitoring water levels, flow rates, water quality and sediment status, and effective management depends on real-time measurements as well as accurate numerical tools.

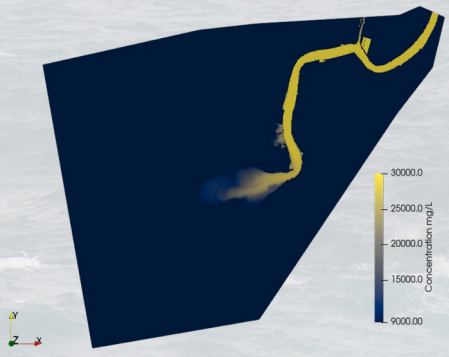
The PhD work is conducted by the research fellow **Laure Sicard** who joined the University of Zaragoza early October 2024. Originated from Reunion Island (France), Laure has a strong background in civic engineering specialising in geology and hydrogeology acquired at the ENSG french engineering school of geology and did a second formation in oceanography with the master Erasmus MUNDUS MER + (which contain the formation of three different universities: University of Bordeaux (France), PIE laboratory from the university of Bilbao (Spain), Southampton University (UK) and AZTI laboratory from Donosti (Spain)).

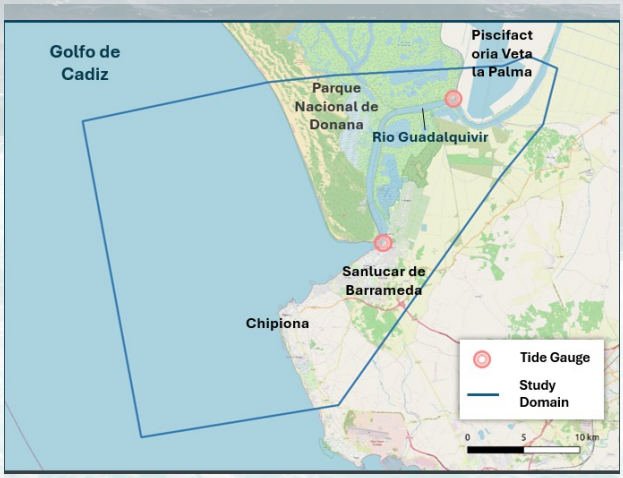
Objectives:

Laure's PhD work lies on the development of computational models enabling the simulation of flow dynamics over erodible beds with variably saturated subsurface flows as well as dry and wet granular flows (see Fig. 13). A numerical 2D coupled surface-subsurface flow model will be extended to handle a moving bed, which affects both the surface and the subsurface computational domain. A vertical multi-layer model is added in order to evaluate the repartition of a scalar within the water column; it could be, in the case of an estuary, sediments, salinity, or even a pollutant. Recent advances in numerical techniques and GPU implementations enable an integrated and comprehensive multi-physics model encompassing and linking all the effects. Experimental data from other studies conducted in and out of the team GHC will be used to justify her work. Notably, a secondment at the university of Ancona will allow her to experiment with a laboratory version of an estuary to assess the possible sediment movements and bed load. Moreover, she will be associated with HYDRONIA, also part of the RESCUER partners, to use her model in a more realistic context

Expected Results:

- Development of a numerical model for simulation of water flows over erodible beds with partially saturated subsurface. The model will allow simultaneous description of river flow, dry and wet granular flows, sediment transport as well as contaminant concentration.
- High performance GPU-based implementation to achieve efficiency and acceptable run times even in large physical domains.
- Evaluation of field data on riverbed features such as sandbars evolving within the river reach and at the river mouth
- Numerical simulation of realistic scenarios arising from consulting work in water management in estuaries.





Guadalquivit study for sediment transport models assessment

Project News

The RESCUER Doctoral Network continues to progress successfully across research, training, and dissemination activities. Over the past months, important milestones have been achieved, reflecting the strong engagement of the consortium and fellows.

Periodic Report Submitted to the European Commission

A major administrative milestone was reached with the successful submission of the **Periodic Report** to the European Commission. This report summarizes the scientific progress, training activities, management actions, communication efforts, and overall achievements of the project during the reporting period. Its submission highlights the dynamic progress of RESCUER and the active collaboration among all partners.

New Scientific Publications

RESCUER partners continue to contribute high-quality scientific outputs in areas directly related to flood modelling, coastal processes, uncertainty quantification, and numerical simulation. Recent journal publications include:

- *Lagrangian Acceleration as a Diagnostic for Wave Breaking in the Nearshore Zone*
  Rosa Maria Vargas-Magaña, Henrik Kalisch, Marc Buckley, Maria Bjørnestad, Tessa Gronemann, Karoline Holand, Jochen Horstmann, Michael Streßer, Sébastien Fromenteau.
  Journal of Geophysical Research: Oceans, 2026

- *Probabilistic assessment of human instability in urban areas exposed to flood events*
  Gabriele Bernardini, Francesco Lagona, Marco Mingione, Matteo Postacchini.
  Scientific Reports, 15, 2025

- *Real-time ocean wave prediction in time domain with autoregression and echo state networks*
  Karoline Holand, Henrik Kalisch
  Frontiers in Marine Science, 11 2024

These publications demonstrate the interdisciplinary strength of the network, spanning numerical methods, coastal hazards, forecasting tools, and human flood safety.

Preparation of Workshop 5a – Zaragoza, Spain

Preparations are underway for **RESCUER Workshop 5a**, which will take place in Zaragoza from **22 to 26 June 2026**.

This training event will focus on Python scientific programming, digital mapping tools, hands-on numerical exercises, and best practices in programming. The workshop will provide fellows with valuable computational skills increasingly required in environmental modeling and research software development.

Upcoming Special Session at HIC 2026

RESCUER will also be represented at the upcoming HIC 2026 in Zaragoza through the special session:

Forecasting coastal and river hazards, flooding and water quality organized by Rita

Carvalho, Claes Eskilsson, Maria Kazolea, and Henrik Kalisch.

This session will highlight recent advances in predictive modeling, environmental resilience, and integrated flood-risk management.

New Project Manager

As of January 22, 2026 Anders Norevik is the new project manager of RESCUER!

Funded by the European Union

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